

### **34. Morphological Operations Using OpenCV – Top Hat Technique**

#### **Aim**

To perform morphological Top Hat transformation on an image using OpenCV.

#### **Procedure**

1. Import the required OpenCV and NumPy libraries.
2. Read the input image.
3. Convert the image into grayscale.
4. Create a structuring element (kernel).
5. Apply the Top Hat transformation using OpenCV.
6. Display the original and processed images.
7. Save the output image.

#### **Program**

```
import cv2

import numpy as np

# Read the input image

img = cv2.imread("input.jpg")

# Convert to grayscale

gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

# Create structuring element

kernel = np.ones((5, 5), np.uint8)
```

```
# Apply Top Hat transformation
```

```
tophat = cv2.morphologyEx(gray, cv2.MORPH_TOPHAT, kernel)
```

```
# Display images
```

```
cv2.imshow("Original Image", img)
```

```
cv2.imshow("Top Hat Image", tophat)
```

```
# Save output image
```

```
cv2.imwrite("tophat_output.jpg", tophat)
```

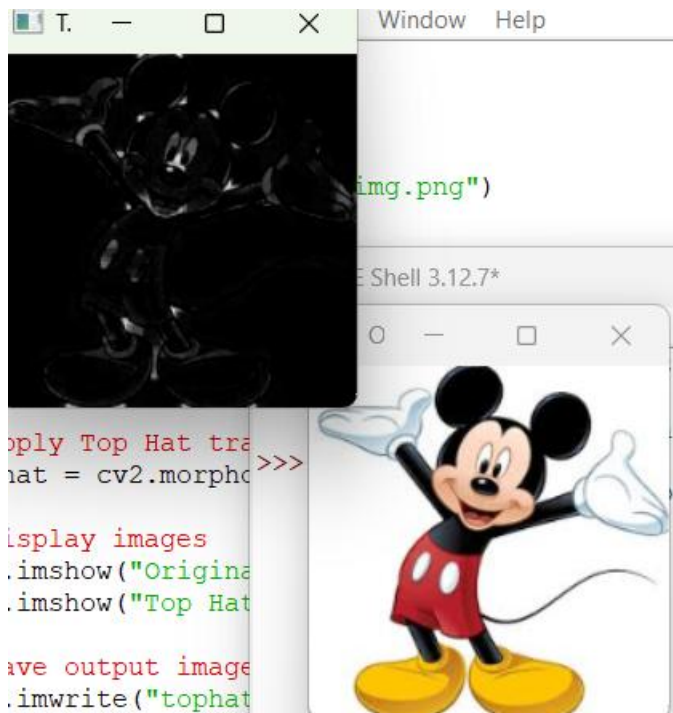
```
cv2.waitKey(0)
```

```
cv2.destroyAllWindows()
```

**Input**



**Output**



## Result

Thus, the morphological Top Hat operation was successfully performed using OpenCV. It helps to extract small bright objects and details from an image background.